Historical Geometry Reconstruction from Surviving Quantum-Informational and Causal Structure

Exploratory Mathematical Specification

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1 Purpose and conceptual shift

We formulate a "reverse Theseus" reconstruction problem. We do not attempt to restore the microscopic state or the original atoms of a historical region. Instead, we ask whether a historical geometric state can be reconstructed, up to declared observational equivalence, from relational information available in a later physical state.

For a historical target label t_0 and later observation state t_1,

$$\mathcal{O}_{t_1} \longrightarrow \widehat{\mathcal{G}}_{t_0}. \tag{1}$$

The label t_0 is bookkeeping for the synthetic benchmark and must not be provided to the reconstruction algorithm as an external physical-time coordinate.

2 Microscopic and observational states

Let

$$\rho(t) \in \mathcal{D}(\mathcal{H}) \tag{2}$$

be a microscopic density operator. We do not require $\widehat{\rho}(t_0) = \rho(t_0)$. Instead introduce an observation map

$$\mathcal{M} : \rho \mapsto \mathcal{O}, \qquad \mathcal{O}(\rho) = \{\mathrm{Tr}(\rho O_k)\}_k. \tag{3}$$

The selected observations may include mutual information, intervention responses, and process statistics.

For an observation family $\mathfrak{O}$, define

$$\rho_a \sim_{\mathfrak{O}} \rho_b \iff \mathcal{M}_{\mathfrak{O}}(\rho_a) = \mathcal{M}_{\mathfrak{O}}(\rho_b). \tag{4}$$

With noise,

$$d_{\mathfrak{O}}(\rho_a, \rho_b) = \sum_k w_k d_k(\mathcal{M}_k(\rho_a), \mathcal{M}_k(\rho_b)). \tag{5}$$

Thus the target is an observational equivalence class, not an exact microscopic state.

3 Relational historical state

Represent a relevant state by

$$\mathcal{R}(t) = (V_t, E_t, \rho_t, \mathcal{P}_t), \tag{6}$$

where V_t are relevant subsystems/events, E_t their relational information, ρ_t quantum state information, and $\mathcal{P}_t$ process and intervention statistics. The elements of V_t need not be the same atoms at different times; components may be destroyed, replaced, merged, split, relocated, or reused.

4 Quantum-information observables

For subsystems A, B,

$$S(A) = -\operatorname{Tr}(\rho_A \log \rho_A), \qquad I(A:B) = S(A) + S(B) - S(AB). \tag{7}$$

The observation vector may contain pairwise and multipartite quantities:

$$\mathcal{I} = \{I(A_i : A_j), I(A_i : A_j : A_k), \dots\}. \tag{8}$$

Pairwise mutual information is not assumed sufficient.

5 Operational causal observables

For a process $\mathcal{E}$ and finite intervention family $\mathfrak{A}$,

$$C_{i\to j} = \max_{a\in\mathfrak{A}} D_{\mathrm{tr}}(\rho_j^{(a)}, \rho_j^{(0)}), \qquad D_{\mathrm{tr}}(\rho, \sigma) = \frac{1}{2}\|\rho - \sigma\|_1. \tag{9}$$

The current implementation is a finite sampled approximation, not the exact operational supremum. Where possible, retain the full intervention-response tensor

$$\mathcal{C} = \{D_{\mathrm{tr}}(\rho_j^{(a)}, \rho_j^{(0)})\}_{i,j,a}. \tag{10}$$

6 Quantum processes and information survival

A fixed-order CPTP process is

$$\mathcal{E}(\rho) = \sum_\alpha K_\alpha \rho K_\alpha^\dagger, \qquad \sum_\alpha K_\alpha^\dagger K_\alpha = I. \tag{11}$$

A synthetic history is

$$\rho_0 \xrightarrow{\mathcal{E}_1} \rho_1 \xrightarrow{\mathcal{E}_2} \cdots \xrightarrow{\mathcal{E}_T} \rho_T. \tag{12}$$

Irreversibility is allowed; reconstruction must not assume that the later state has a physical inverse.

7 Historical geometry

Let

$$\mathcal{G}_{t_0} = (M_{t_0}, g_{t_0}) \tag{13}$$

with Lorentzian signature $(-,+,+,+)$. For suitable events,

$$s_{ij} = g_{\mu\nu}\Delta x^\mu \Delta x^\nu, \tag{14}$$

with

$$\chi_{ij} = \begin{cases} \text{timelike}, & s_{ij} < 0, \\ \text{null}, & s_{ij} = 0, \\ \text{spacelike}, & s_{ij} > 0. \end{cases} \tag{15}$$

The target is geometric equivalence, not a particular coordinate chart.

8 Inverse problem

Let the forward physical construction be

$$\Phi : \mathcal{G} \longrightarrow \mathcal{O}. \tag{16}$$

The central question is whether a reconstruction map exists:

$$\mathcal{R} : \mathcal{O}_{t_1} \longrightarrow [\mathcal{G}_{t_0}] \tag{17}$$

such that

$$\mathcal{R}(\mathcal{O}_{t_1}) \approx \mathcal{G}_{t_0}. \tag{18}$$

Three failure modes must be separated:

1. **Information insufficiency:** distinct geometries produce indistinguishable observables.
2. **Representation insufficiency:** observables are informative but the reconstruction family cannot represent the target.
3. **Estimator/optimization insufficiency:** representation is adequate but sampling, optimization, or numerical procedures fail.

9 Geometry rather than coordinates

Coordinate error is only a diagnostic. The primary reconstruction target is

$$\widehat{g}_{\mu\nu}(x) \sim g_{\mu\nu}(x), \tag{19}$$

up to the declared coordinate gauge. Evaluation should include causal relations, invariant intervals, Lorentzian signature, non-degeneracy, and curvature invariants for curved worlds.

10 Synthetic historical worlds

Construct

$$\mathcal{G}_0 \to \mathcal{G}_1 \to \cdots \to \mathcal{G}_T \tag{20}$$

together with an associated information process

$$\rho_0 \to \rho_1 \to \cdots \to \rho_T. \tag{21}$$

The reconstruction receives later-state relational observables, not hidden historical geometry.

The benchmark must include:

1. a randomized/null control;
2. a positive-control construction in which the physical process is explicitly coupled to the target relational structure;
3. multiple historical geometries and unseen held-out geometries;
4. controlled information loss and observation noise.

The positive control is essential: the existing independent-observable experiment only establishes that unrelated observables do not robustly reconstruct geometry.

11 Information-retention experiment

Introduce $\eta \in [0,1]$ as a controlled information-retention parameter and let $\mathcal{O}_{t_1}^{(\eta)}$ be the retained observations. Define

$$F_{\mathrm{geom}}(\eta) = \mathrm{Score}(\widehat{\mathcal{G}}_{t_0}^{(\eta)}, \mathcal{G}_{t_0}). \tag{22}$$

Measure reconstruction quality as relational information is progressively erased. This is a benchmark quantity, not a proposed physical law.

12 Reconstruction objective

A generic baseline objective is

$$\mathcal{L} = \lambda_c \mathcal{L}_{\mathrm{causal}} + \lambda_s \mathcal{L}_{\mathrm{interval}} + \lambda_r \mathcal{L}_{\mathrm{curvature}} + \lambda_{\mathrm{reg}} \mathcal{L}_{\mathrm{reg}}. \tag{23}$$

For example,

$$\mathcal{L}_{\mathrm{causal}} = \mathrm{BCE}(\widehat{C}, C_{\mathrm{GT}}), \tag{24}$$

and

$$\mathcal{L}_{\mathrm{interval}} = \sum_{i<j} w_{ij} |\widehat{s}_{ij} - s_{ij}|^2. \tag{25}$$

Curvature losses are used only for target families where curvature is a declared observable. No speculative correction tensor is needed in the baseline.

13 Identifiability

If

$$\Phi(\mathcal{G}_1) = \Phi(\mathcal{G}_2), \qquad \mathcal{G}_1 \not\sim \mathcal{G}_2, \tag{26}$$

then the observation family is insufficient to uniquely identify geometry. The correct output is an equivalence class, not forced unique recovery.

14 Relation to the current implementation

Retain the existing finite infrastructure: density matrices, constraint and relational-clock toy model, mutual information, finite causal interventions, fixed-order Kraus processes, Lorentzian geometry, GR tensor diagnostics, ground-truth worlds, held-out evaluation, coordinate tests, and reproducibility.

The current independent-observable failure remains a valid null baseline:

$$\text{unrelated observables} \not\Rightarrow \text{geometry}. \tag{27}$$

The new experiment instead tests

$$\boxed{\text{historically coupled relational information} \longrightarrow \text{historical geometry.}} \tag{28}$$

15 Frozen speculative sectors

Throughout the baseline experiments,

$$Q = 0, \qquad \Delta_{\mu\nu} = 0. \tag{29}$$

No speculative modification is introduced to rescue reconstruction failure.

16 Allowed scientific claims

A positive result may support only a statement that a specified family of historical geometries is reconstructible, to quantified accuracy and under specified observational equivalence, from declared later-state quantum-information and causal observables. It does not establish time travel, exact microscopic recovery, emergent spacetime in nature, or new fundamental physics.

A negative result must be classified, where possible, as information insufficiency, non-identifiability, representation insufficiency, or optimization failure.

17 Baba Novac interpretation

The conceptual target is

$$\boxed{\texttt{BabaNovac@2026} \longrightarrow \texttt{BabaNovac@1987}}. \tag{30}$$

This is not literal reversal of atoms. It means reconstruction of a historical observational equivalence class from relational information that survives in the later physical state.